Quentin Garrido

AI Researcher Website : garridoq.com

Education

- **Meta - FAIR, Université Gustave Eiffel** Paris, France
  *PhD in computer science; Co-advised by **Yann LeCun** and **Laurent Najman**.* *Mar 2022 – May 2025*
- **École Normale Supérieure Paris-Saclay** Gif-sur-Yvette, France
  *MSc in Mathematics, Computer Vision and Machine Learning (MVA); **highest honors*** *2020 – 2021*
- **ESIEE Paris Engineering School** Noisy-Le-Grand, France
  *MSc in Computer Science; **top of the class*** *2016 – 2021*

Work Experience

- **AMI Labs** Paris, France
  Member of Technical Staff *Jun 2026 – Now*
  - **AI for the physical world**: Building the next generation of AI systems that can understand the physical world, as well as plan and reason in it.
- **Meta - Fundamental AI Research (FAIR)** Paris, France
  Research Scientist *Mar 2025 – Jun 2026*
  - **Joint Embedding Predictive Architectures**: Part of the JEPA team, I worked on topics pertaining to self-supervised learning and world models. I contributed to V-JEPA 2, a state of the art video encoder. I explored the learning of world models on Youtube scale videos using latent action models, providing the first evidence that such models can be trained at scale and used effectively.

  Research Scientist Assistant, Industrial PhD program *Mar 2022 – Mar 2025*
  - **Self-supervised learning**: Researched various aspects of self-supervised learning and its applications: theory, label-free evaluation, equivariant representation learning, application to physics, links to world modeling, study of physics understanding in video models. Led to 12 publications, including 6 (co-)first authored ones.
  - **Supervision**: Co-advised by *Yann LeCun* and *Laurent Najman*.

  Research intern *May 2021 – Sept 2021*
  - **Self-Supervised Learning**: Developed a self-supervised approach to learn image representations that contain invariant and equivariant information with respect to a set of transformations. The method relies on the optimisation of regularized latent variables to learn the transformation in representation space.
  - **Supervision**: Supervised by *Yann LeCun* and *Laurent Najman*.
- **Heidelberg University - HCI - Image Analysis and Learning lab** Heidelberg, Germany
  Research intern *May 2020 – Aug 2020*
  - **Machine learning on graphs**: Developed a new autoencoder based dimensionality reduction method which aims to preserve the hierarchical properties present in the data.
    Combined deep learning methods and graph based techniques such as a density based Minimum Spanning Tree.
    Applied and evaluated the method on single-cell transcriptomics data to visualize cellular differentiation.
  - **Supervision**: Supervised by *Laurent Najman* and *Fred Hamprecht*.

Awards

- **ICLR 2023 Outstanding Paper Honorable Mention (9/1574 papers)**:
  On the duality between contrastive and non contrastive self-supervised learning
- **Ian Lawson Van Toch Memorial Award for Outstanding Student Paper 2022 (1/48 papers)**:
  Visualizing hierarchies in scRNA-seq data using a density tree-biased autoencoder

Publications

First authored publications

- **Quentin Garrido**, Tushar Nagarajan, Basile Terver, Nicolas Ballas, Yann LeCun, and Michael Rabbat. Learning Latent Action World Models In The Wild. *International Conference on Machine Learning*, 2026.
- **Quentin Garrido**, Nicolas Ballas, Mahmoud Assran, Adrien Bardes, Laurent Najman, Michael Rabbat, Emmanuel Dupoux, and Yann LeCun. Intuitive physics understanding emerges from self-supervised pretraining on natural videos. *Under review at PNAS*, 2025.
- **Quentin Garrido**, Mahmoud Assran, Nicolas Ballas, Adrien Bardes, Laurent Najman, and Yann LeCun. Learning and leveraging world models in visual representation learning. *arXiv preprint*, 2024.
- **Quentin Garrido***, Grégoire Mialon*, Hannah Lawrence, Danyal Rehman, Yann LeCun, and Bobak Kiani. Self-supervised learning with lie symmetries for partial differential equations. *Advances in Neural Information Processing Systems*, 36:28973–29004, 2023.
- **Quentin Garrido**, Randall Balestriero, Laurent Najman, and Yann Lecun. Rankme: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *International conference on machine learning*, 2023. **(Oral Presentation)**
- **Quentin Garrido**, Laurent Najman, and Yann Lecun. Self-supervised learning of split invariant equivariant representations. In *International Conference on Machine Learning*, 2023.
- **Quentin Garrido**, Yubei Chen, Adrien Bardes, Laurent Najman, and Yann LeCun. On the duality between contrastive and non-contrastive self-supervised learning. In *The Eleventh International Conference on Learning Representations*, 2023. **(Outstanding Paper Honorable Mention)**
- **Quentin Garrido**, Sebastian Damrich, Alexander Jäger, Dario Cerletti, Manfred Claassen, Laurent Najman, and Fred A Hamprecht. Visualizing hierarchies in scrna-seq data using a density tree-biased autoencoder. *Bioinformatics*, 38(Supplement_1):i316–i324, 2022 **(Ian Lawson Van Toch Memorial Award for Outstanding Student Paper)**.

Other publications

- Sonia Joseph, **Quentin Garrido**, Randall Balestriero, Matthew Kowal, Thomas Fel, Shahab Bakhtiari, Blake Richards, Michael Rabbat. Interpreting physics in video world models. *International Conference on Machine Learning*, 2026.
- Basile Terver, Randall Balestriero, Megi Dervishi, David Fan, **Quentin Garrido**, Tushar Nagarajan, Koustuv Sinha, Wancong Zhang, Mike Rabbat, Yann LeCun, and Amir Bar. A lightweight library for energy-based joint-embedding predictive architectures. *ICLR Workshop on World Models*, 2026.
- Mido Assran, Adrien Bardes, David Fan, **Quentin Garrido**, Russell Howes, Matthew Muckley, Ammar Rizvi et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint*, 2025.
- Benno Krojer, Mojtaba Komeili, Candace Ross, **Quentin Garrido**, Koustuv Sinha, Nicolas Ballas, and Mahmoud Assran. A shortcut-aware video-qa benchmark for physical understanding via minimal video pairs. *Transactions on Machine Learning Research*, 2025.
- Florian Bordes, **Quentin Garrido**, Justine T. Kao, Adina Williams, Michael Rabbat, and Emmanuel Dupoux. IntPhys 2: Benchmarking intuitive physics understanding in complex synthetic environments. *arXiv preprint*, 2025.
- Adrien Bardes, **Quentin Garrido**, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024.

- Haider Al-Tahan, **Quentin Garrido**, Randall Balestriero, Diane Bouchacourt, Caner Hazirbas, and Mark Ibrahim Scaling vision-language models does not improve relational understanding: The right learning objective helps. *NeurIPS datasets and benchmarks*, 2024.
- Florian Bordes, Richard Yuanzhe Pang, Anurag Ajay, Alexander C Li, Adrien Bardes, Suzanne Petryk, Oscar Mañas, Zhiqiu Lin, Anas Mahmoud, Bargav Jayaraman, et al. An introduction to vision-language modeling. *arXiv preprint*, 2024.
- Randall Balestriero, Mark Ibrahim, Vlad Sobal, Ari Morcos, Shashank Shekhar, Tom Goldstein, Florian Bordes, Adrien Bardes, Gregoire Mialon, Yuandong Tian, et al. A cookbook of self-supervised learning. *arXiv preprint*, 2023.
- Florian Bordes, Randall Balestriero, **Quentin Garrido**, Adrien Bardes, and Pascal Vincent. Guillotine Regularization: Why removing layers is needed to improve generalization in Self-Supervised Learning. *Transactions on Machine Learning Research*, 2022.
- Mark Ibrahim, **Quentin Garrido**, Ari Morcos, and Diane Bouchacourt. The robustness limits of sota vision models to natural variation. *Transactions on Machine Learning Research*, 2022.

Service and Teaching

Reviewer for ECCV (2022,2024), NeurIPS (2024, 2025), ICCV (2023), ICLR (2025), International Conference on Machine Learning (2026), Self-Supervised Learning workshop at NeurIPS (2022/2023/2024), TMLR.
Teaching Assistant for the Master level "Algorithm Design" class at ESIEE Paris (2021/2022/2023).

Invited talks

- **Physical AI Tech Week (Korea), (14/10/2026)**: World Models : From Future Prediction to Real-World Understanding
- **Korea's National IT Industry Promotion Agency (NIPA), (10/07/2026)**: Predicting the future of the world before acting in it with world models
- **Samsung Research - AI Center (06/07/2026)**: Towards AI for the physical world: World Models and Latent Prediction
- **LIX Seminar (École Polytechnique) (21/05/2026)**: Towards AI for the physical world: latent prediction and world models
- **UNAITE Seminar (11/02/2026)**: World Models and Latent Prediction
- **Guest Lecture at Brown University (30/10/2024)**: Self-supervised learning
- **IMAGINE Seminar (29/05/2024)**: Understanding Self-Supervised Learning to go beyond invariant image representations
- **GTTI Seminar (Centre Borelli) (29/02/2024)**: Self-Supervised Learning: From learning representations to learning world models
- **AI4Science Talks (30/11/2023)**: Self-Supervised Learning with Lie Symmetries for Partial Differential Equations
- **NYU CILVR Seminar (26/10/2023)**: Self-Supervised Learning: Going beyond images
- **Extrality Seminar (04/10/2023)**: Self-Supervised Learning with Lie Symmetries for PDEs
- **Thales Research Seminar (21/07/2023)**: On the duality between contrastive and non contrastive self-supervised learning